TASK 5 REPORT

ELK Stack SIEM Monitoring and Detection

Report	Task 5 Report: ELK Stack SIEM Monitoring and Detection
SIEM Solution	Elastic Stack (Elasticsearch and Kibana)
Elasticsearch Version	9.4.2
Operating System	Windows Operating System

Objective

The objective of this task was to configure and use the Elastic Stack (Elasticsearch and Kibana) as a Security Information and Event Management (SIEM) solution. The task involved ingesting simulated security logs, creating detection rules, generating alerts, and building an interactive dashboard for monitoring security events.

1. Tools Used

- Elasticsearch 9.4.2
- Kibana
- Elastic Security
- Kibana Dev Tools
- Kibana Lens
- Windows Operating System

2. Data Collection

A custom Elasticsearch index named attack-logs was created to store simulated security events.

The dataset included multiple attack scenarios such as:

- Login Events
- DNS Events
- PowerShell Events
- Privilege Escalation Events

Each document contained important fields including:

Field	Field
@timestamp	event_type
source_ip	username
process_name	command_line
severity	status
dns_query	powershell_command

A total of 39 simulated security events were successfully indexed into Elasticsearch.

3. Detection Rules

The following custom detection rules were created in Elastic Security:

- Credential Stuffing Detection
- DNS Tunnelling Detection
- PowerShell Exploitation Detection
- Privilege Escalation Detection

Each rule was configured to monitor the attack-logs index using Kibana Query Language (KQL) and executed on a scheduled interval.

4. Alert Generation

The detection rules successfully generated alerts based on the simulated attack data.

The generated alerts included:

- Credential Stuffing Alerts
- DNS Tunnelling Alerts
- PowerShell Exploitation Alerts
- Privilege Escalation Alerts

The alerts were categorized into different severity levels:

- High
- Medium
- Low

A total of 25 alerts were generated during testing, confirming that the detection rules operated successfully.

5. Dashboard Development

A Security Operations Center (SOC) dashboard was created using Kibana Lens to provide a visual representation of the collected security events.

The dashboard contains the following visualizations:

- Privilege Escalation Events
- Top Event Types
- Top Source IP Addresses
- Events by Severity
- Login Events Over Time
- DNS Events Over Time
- PowerShell Events Over Time
- Attack Timeline

These visualizations provide an overview of attack activity, event distribution, severity levels, source IP addresses, and the timeline of security events.

6. Results

The ELK Stack SIEM successfully achieved the following:

- Successfully ingested simulated security events into Elasticsearch.
- Created multiple custom detection rules.
- Generated alerts based on simulated attack scenarios.
- Classified alerts by severity.
- Built an interactive SOC dashboard for monitoring security events.
- Verified alerts through the Elastic Security interface.

The completed dashboard enables efficient monitoring and analysis of simulated cyber attacks.

7. Screenshots

The following screenshots are included as evidence of successful implementation:

- Discover page showing indexed attack logs.
- Detection Rules page.
- Alerts page displaying generated alerts.
- SOC Dashboard containing all visualizations.
- Dev Tools query output (optional).

8. Conclusion

This task demonstrated the implementation of a functional SIEM environment using the Elastic Stack. Simulated security events were successfully indexed into Elasticsearch, monitored using custom detection rules, and visualized through Kibana dashboards. Elastic Security generated alerts for multiple attack scenarios, validating the effectiveness of the configured rules. The completed dashboard provides centralized visibility into security events and demonstrates core SOC monitoring capabilities, including threat detection, alert analysis, and security visualization.